

Week 4 Discussion Solutions

Thermodynamics: Processes, Heat Engines, Second Law, Entropy

Introduction Exercise: Key Equations

1. (a) $U = \frac{d}{2}NkT$
- (b) $PV = NkT = nRT$
- (c) $dU = dQ - dW$
- (d) $dW = P dV$
- (e) $dQ = mc dT = C dT$
- (f) $dQ = L dm$
- (g) $\eta = \frac{W}{Q_1}$
- (h) $dS = \frac{\delta Q_{\text{rev}}}{T}$
- (i) $\frac{dQ}{dt} = -kA \frac{dT}{dx}$

Part I: Heat Engines

Exercise 1

- (a) We can solve for the volume at point 3 with the fact that $PV^\gamma = \text{const}$ throughout an adiabatic process. $\gamma = \frac{5}{3}$ for a monatomic ideal gas, so

$$2P_0(3V_0)^{5/3} = P_0V^{5/3} \implies V = 2^{3/5}(3V_0). \quad (1)$$

Using the fact that $E_{\text{int}} = \frac{3}{2}NkT = \frac{3}{2}PV$ and the first law of thermodynamics, we can populate this table:

	ΔE_{int}	W	Q
$1 \rightarrow 2$	$\frac{3}{2}(6P_0V_0) - \frac{3}{2}(2P_0V_0) = 6P_0V_0$	$(2P_0)(2V_0) = 4P_0V_0$	$10P_0V_0$
$2 \rightarrow 3$	$\frac{3}{2}(2^{3/5}3P_0V_0) - \frac{3}{2}(6P_0V_0) \approx -2.18P_0V_0$	$\approx 2.18P_0V_0$	0
$3 \rightarrow 4$	$\frac{3}{2}(P_0V_0) - \frac{3}{2}(2^{3/5}3P_0V_0) \approx -5.32P_0V_0$	$-(2^{3/5}3 - 1)P_0V_0 \approx -3.55P_0V_0$	$\approx -8.87P_0V_0$
$4 \rightarrow 1$	$\frac{3}{2}(2P_0V_0) - \frac{3}{2}(P_0V_0) = \frac{3}{2}P_0V_0$	0	$\frac{3}{2}P_0V_0$

The net work done over the whole cycle is $W_{\text{tot}} = (4 + 2.18 - 3.55)P_0V_0 \approx 2.63P_0V_0$, and the overall heat input is $(10 + 1.5)P_0V_0 = 11.5P_0V_0$. Thus, the efficiency of this heat engine is $\eta = \frac{W}{Q} \approx 0.23$.

- (b) The highest temperature is achieved at point 2, and the lowest temperature is at point 4. By the ideal gas law,

$$(2P_0)(3V_0) = NkT_h \implies T_h = \frac{6P_0V_0}{Nk}, \quad (2)$$

$$P_0V_0 = NkT_c \implies T_c = \frac{P_0V_0}{Nk}. \quad (3)$$

- (c) Carnot efficiency is given by

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h} = 1 - \frac{1}{6} = \frac{5}{6} \approx 0.83. \quad (4)$$

- (d) The engine we analyzed here is indeed less efficient than the Carnot engine, consistent with the fact that the Carnot engine is the most efficient engine possible.

Exercise 2

1. We know $\eta_A = \frac{W_A}{Q_1^A}$ by definition, so $W_A = \eta_A Q_1^A$. By conservation of energy, $Q_1^A = W_A + Q_2^A$, so

$$Q_2^A = Q_1^A - W_A = Q_1^A (1 - \eta_A).$$

2. By definition, $\eta_B = \frac{W_B}{Q_1^B}$, so

$$W_B = \eta_B Q_1^B = \eta_B Q_2^A = \eta_B (1 - \eta_A) Q_1^A.$$

Similarly,

$$Q_2^B = Q_1^B - W_B = (1 - \eta_B) Q_1^B = (1 - \eta_B) (1 - \eta_A) Q_1^A.$$

3. The total work output is

$$W = W_A + W_B = \eta_A Q_1^A + \eta_B (1 - \eta_A) Q_1^A = (\eta_A + \eta_B (1 - \eta_A)) Q_1^A.$$

4. The net efficiency of the combined system is

$$\eta_{\text{tot}} = \frac{W}{Q_1^A} = \frac{Q_1^A - Q_2^B}{Q_1^A} = 1 - (1 - \eta_A)(1 - \eta_B).$$

5. Since $\eta_A < 1$ and $\eta_B < 1$, we have $0 < (1 - \eta_A) < 1$ and $0 < (1 - \eta_B) < 1$. Therefore $0 < (1 - \eta_A)(1 - \eta_B) < 1$, which implies

$$\eta_{\text{tot}} = 1 - (1 - \eta_A)(1 - \eta_B) < 1.$$

6. (If both engines are Carnot.) If $\eta_A = 1 - \frac{T_M}{T_H}$ and $\eta_B = 1 - \frac{T_C}{T_M}$, then

$$\eta_{\text{tot}} = 1 - (1 - \eta_A)(1 - \eta_B) = 1 - \left(\frac{T_M}{T_H}\right) \left(\frac{T_C}{T_M}\right) = 1 - \frac{T_C}{T_H},$$

which is the Carnot efficiency between T_C and T_H .

Part II: Second Law of Thermodynamics

Exercise 1

- (a) Because the system is isolated, we know that $\Delta E = 0$. Thus,

$$\begin{aligned} 0 &= \Delta E_{\text{ice}} + \Delta E_{\text{melted water}} + M_{\text{ice}}L_f + \Delta E_{\text{lead}} \\ &= M_{\text{ice}}c_{\text{ice}}(T_m - T_{\text{ice}}) + M_{\text{ice}}c_{\text{water}}(T_f - T_m) + M_{\text{ice}}L_f + M_L c_L (T_f - T_L), \end{aligned}$$

where T_m is the melting temperature of ice, and T_L is the initial temperature of the lead bowl.

- (b) For the lead bowl,

$$dS_L = \frac{M_L c_L dT}{T} \quad \Rightarrow \quad \Delta S_L = M_L c_L \ln\left(\frac{T_f}{T_L}\right).$$

- (c) For the water/ice (using a reversible path),

$$\Delta S_{\text{H}_2\text{O}} = M_{\text{ice}}c_{\text{ice}} \ln\left(\frac{T_m}{T_{\text{ice}}}\right) + \frac{M_{\text{ice}}L_f}{T_m} + M_{\text{ice}}c_{\text{water}} \ln\left(\frac{T_f}{T_m}\right).$$

- (d) For the combined system, the entropy change is positive: $\Delta S_{\text{sys}} > 0$ (second law; the process is irreversible).

- (e)

$$\Delta S_{\text{sys}} = \Delta S_L + \Delta S_{\text{H}_2\text{O}}.$$

Exercise 2

First, we think physically about what is happening. In Process 1, the gases mix by a “free expansion” into the full container; this is irreversible. In Process 2, each gas undergoes a reversible isothermal expansion while in contact with a reservoir at temperature T .

From gas A’s perspective, gas B and the membrane permeable to A do not affect the equation of state; we can treat the entropy changes of gases A and B independently and add them at the end. A schematic of the transformation is shown in Fig. 1.

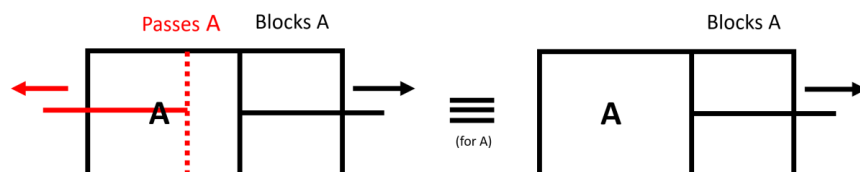


Figure 1: Schematic for the reversible (membrane) mixing process.

- (a) (Process 2.) For gas A, the process is isothermal and reversible, so

$$\Delta S_A = \int \frac{\delta Q_{\text{rev}}}{T} = \frac{1}{T} \int \delta Q, \quad (5)$$

since T is constant. For an ideal gas in an isothermal process, $\Delta U = 0$, so by the first law $\delta Q = \delta W = P dV$. Therefore

$$\Delta S_A = \frac{1}{T} \int_{V_A}^{V_A+V_B} P dV = \frac{1}{T} \int_{V_A}^{V_A+V_B} \frac{n_A R T}{V} dV = n_A R \ln \left(\frac{V_A + V_B}{V_A} \right). \quad (6)$$

Similarly,

$$\Delta S_B = n_B R \ln \left(\frac{V_A + V_B}{V_B} \right). \quad (7)$$

Hence, for Process 2,

$$\Delta S_{\text{container}} = n_A R \ln \left(\frac{V_A + V_B}{V_A} \right) + n_B R \ln \left(\frac{V_A + V_B}{V_B} \right). \quad (8)$$

- (b) (Process 1.) This process is irreversible, so one should not compute ΔS via $\int \delta Q/T$ along the irreversible path. Instead, use any reversible path between the same initial and final equilibrium states. Since Process 2 has the same initial and final states as Process 1, and entropy is a state function,

$$\Delta S_{\text{container}} = n_A R \ln \left(\frac{V_A + V_B}{V_A} \right) + n_B R \ln \left(\frac{V_A + V_B}{V_B} \right). \quad (9)$$

- (c) (Reservoir entropy in Process 2.) Process 2 is reversible and isothermal, so the total entropy change of the universe is zero. Therefore

$$\Delta S_{\text{res}} = -\Delta S_{\text{container}} = -n_A R \ln \left(\frac{V_A + V_B}{V_A} \right) - n_B R \ln \left(\frac{V_A + V_B}{V_B} \right). \quad (10)$$

As a final comment: in Process 1 the container is adiabatically isolated, so $\Delta S_{\text{res}} = 0$ and the entropy change of the universe equals $\Delta S_{\text{container}} > 0$, confirming irreversibility.

Exercise 3

- (a) During process A, the volume of the gas is held constant, so $V_2 = V_1$. The problem states that point 2 is at four times the initial pressure, so $P_2 = 4P_1$.

From the adiabatic gas law applied to process B,

$$P_3 V_3^\gamma = P_2 V_2^\gamma = 4P_1 V_1^\gamma.$$

Since process C is isothermal back to the initial state, $P_3 V_3 = P_1 V_1$. Plugging this into the adiabatic relation gives

$$\left(\frac{P_1 V_1}{V_3} \right) V_3^\gamma = 4P_1 V_1^\gamma \quad \Rightarrow \quad V_3^{\gamma-1} = 4V_1^{\gamma-1} \quad \Rightarrow \quad V_3 = 4^{\frac{1}{\gamma-1}} V_1.$$

For a diatomic ideal gas (no vibrational degrees of freedom), $\gamma = \frac{7}{5}$, so

$$V_3 = 4^{5/2}V_1 = 32V_1, \quad P_3 = \frac{P_1V_1}{V_3} = \frac{P_1}{32}.$$

- (b) We know: step A is isochoric ($W_A = 0$), step B is adiabatic ($Q_B = 0$), and step C is isothermal ($\Delta E_C = 0$).

Using the first law $\Delta E = Q - W$ for step A, $Q_A = \Delta E_A$. For the whole cycle, $\Delta E_{\text{tot}} = 0$, so $\Delta E_B = -\Delta E_A$. For step B, then, $W_B = -\Delta E_B = \Delta E_A$.

For a diatomic ideal gas, $C_V = \frac{5}{2}R$, so

$$\Delta E_A = nC_V(T_2 - T_1) = \frac{5}{2}nR(T_2 - T_1).$$

From the ideal gas law with $V_2 = V_1$ and $P_2 = 4P_1$, we have $T_2 = 4T_1$, hence

$$\Delta E_A = \frac{15}{2}nRT_1.$$

Therefore,

$$\begin{aligned} Q_A &= \frac{15}{2}nRT_1, & W_A &= 0, & \Delta E_A &= \frac{15}{2}nRT_1, \\ Q_B &= 0, & W_B &= \frac{15}{2}nRT_1, & \Delta E_B &= -\frac{15}{2}nRT_1. \end{aligned}$$

For step C (isothermal at T_1),

$$W_C = \int_{V_3}^{V_1} P dV = \int_{32V_1}^{V_1} \frac{nRT_1}{V} dV = -nRT_1 \ln 32.$$

Since $\Delta E_C = 0$, we have $Q_C = W_C = -nRT_1 \ln 32$.

Using $n = 1$ mole, the results are:

	ΔE	Q	W
Process A	$\frac{15}{2}RT_1$	$\frac{15}{2}RT_1$	0
Process B	$-\frac{15}{2}RT_1$	0	$\frac{15}{2}RT_1$
Process C	0	$-RT_1 \ln 32$	$-RT_1 \ln 32$

Exercise 4

Suppose two objects have identical mass m and specific heat c , with initial temperatures $T_H > T_L$. They are put in thermal contact and reach equilibrium T_f .

Energy conservation (no work, no heat exchange with the environment) gives

$$mc(T_f - T_L) = mc(T_H - T_f) \quad \Rightarrow \quad T_f = \frac{T_H + T_L}{2}.$$

To compute entropy changes, use a reversible path for each object:

$$\begin{aligned}\Delta S_H &= \int_{T_H}^{T_f} \frac{mc dT}{T} = mc \ln\left(\frac{T_f}{T_H}\right), \\ \Delta S_L &= \int_{T_L}^{T_f} \frac{mc dT}{T} = mc \ln\left(\frac{T_f}{T_L}\right).\end{aligned}$$

Therefore,

$$\Delta S_{\text{tot}} = \Delta S_H + \Delta S_L = mc \left[\ln\left(\frac{T_f}{T_H}\right) + \ln\left(\frac{T_f}{T_L}\right) \right] = mc \ln\left(\frac{T_f^2}{T_H T_L}\right).$$

Since $T_f = \frac{T_H + T_L}{2} > \sqrt{T_H T_L}$ for $T_H \neq T_L$ (AM–GM inequality), we have $T_f^2 > T_H T_L$, so $\Delta S_{\text{tot}} > 0$. Hence the process is irreversible.